

Probability

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$$1) \text{ Probability} = \frac{\text{Favourable no. of cases (m)}}{\text{Exhaustive no. of cases (n)}}$$

Exhaustive cases (n) = whole total number
Favourable cases (m) = sub-total number

2) Combination (selection):

$${}^n C_r = \frac{n!}{(n-r)! r!}$$

where,

! = Factorial

$$1! = 1$$

$$2! = 2 \times 1$$

$$3! = 3 \times 2 \times 1$$

$$4! = 4 \times 3 \times 2 \times 1$$

Q) A problem is given to three students A, B and C whose chances of solving the problem ~~and~~ are 2:3:5. If all of them try to solve the problem, what is the probability that

- i) none of them will solve the problem
- ii) the problem will be solved
- iii) all of them will solve the problem
- iv) one of them will solve the problem
- v) two of them will solve the problem

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Let A = event of solving problem by student 'A'

B = event of solving problem by student 'B'

C = event of solving problem by student 'C'

Then, $P(A) = \frac{2}{10} = 0.2 \quad \therefore P(\bar{A}) = 1 - 0.2 = 0.8$

$$P(B) = \frac{3}{10} = 0.3 \quad \therefore P(\bar{B}) = 1 - 0.3 = 0.7$$

$$P(C) = \frac{5}{10} = 0.5 \quad \therefore P(\bar{C}) = 1 - 0.5 = 0.5$$

Then,

(i) Probability that none of them will solve the problem

$$\begin{aligned} &= P(\bar{A} \cap \bar{B} \cap \bar{C}) \\ &= P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) \\ &= 0.8 \times 0.7 \times 0.5 = 0.28 \end{aligned}$$

ii) Probability that problem will be solved,

$$\begin{aligned} &= P(A \cup B \cup C) \\ &= P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(A \cap C) \\ &\quad + P(A \cap B \cap C) \\ &= P(A) + P(B) + P(C) - P(A) \times P(B) - P(B) \times P(C) - P(A) \times P(C) \\ &\quad + P(A) \times P(B) \times P(C) \\ &= 0.2 + 0.3 + 0.5 - 0.2 \times 0.3 - 0.3 \times 0.5 - 0.2 \times 0.5 + 0.2 \times 0.3 \times 0.5 \\ &= 0.2 + 0.3 + 0.5 - 0.06 - 0.15 - 0.10 + 0.05 \\ &= 0.72 \end{aligned}$$

Alternative method

Probability that problem will be solved

$$= 1 - P(\bar{A} \cap \bar{B} \cap \bar{C})$$

$$= 1 - P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C})$$

gurukul $= 1 - 0.8 \times 0.7 \times 0.5 = 0.72$

Poisson Distribution:

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$$P(X=r) = \frac{e^{-\lambda} \lambda^r}{r!}$$

$$r = 0, 1, 2, \dots, \infty$$

Note: ~~ii~~

$$P(\text{At least one}) + P(\text{neither}) = 1$$

$$\therefore P(\text{At least one}) = 1 - P(\text{neither})$$

iii) Probability that all of them will be solved

$$\begin{aligned} &= P(A \cap B \cap C) \\ &= P(A) \times P(B) \times P(C) \\ &= 0.2 \times 0.3 \times 0.5 \\ &= 0.03 \end{aligned}$$

iv) Probability that one of them will be solved

$$\begin{aligned} &= P(A \cap \bar{B} \cap \bar{C}) \text{ or } (\bar{A} \cap \bar{B} \cap \bar{C}) \text{ or } (\bar{A} \cap \bar{B} \cap C) \\ &= P(A \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap \bar{C}) + P(\bar{A} \cap \bar{B} \cap C) \\ &= P(A) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(\bar{B}) \cdot P(\bar{C}) + P(\bar{A}) \cdot P(\bar{B}) \cdot P(C) \\ &= 0.2 \times 0.7 \times 0.5 + 0.8 \times 0.3 \times 0.5 + 0.8 \times 0.7 \times 0.5 \\ &= 0.07 + 0.12 + 0.28 \\ &= 0.47 \end{aligned}$$

v) Probability that two of them will be solved

$$\begin{aligned} &= P(A \cap B \cap \bar{C}) \text{ or } (A \cap \bar{B} \cap C) \text{ or } (\bar{A} \cap B \cap C) \\ &= P(A \cap B \cap \bar{C}) + P(A \cap \bar{B} \cap C) + P(\bar{A} \cap B \cap C) \\ &= P(A) \cdot P(B) \cdot P(\bar{C}) + P(A) \cdot P(\bar{B}) \cdot P(C) + P(\bar{A}) \cdot P(B) \cdot P(C) \end{aligned}$$

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$$\begin{aligned}
 &= 0.2 \times 0.3 \times 0.5 + 0.2 \times 0.7 \times 0.5 + 0.8 \times 0.3 \times 0.5 \\
 &= 0.03 + 0.07 + 0.12 \\
 &= 0.22
 \end{aligned}$$

* Bayes's Theorem

Let any event 'A' occurs in connection with any one of the two events E_1 and E_2 . Then probability that event A occurs due to Event E_1 is denoted by ~~$P(E_1)$~~ $P(E_1/A)$ and is given by,

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)}$$

Similarly, probability that event A occurs due to event E_2 is:

$$P(E_2/A) = \frac{P(E_2)P(A/E_2)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2)}$$

For three events E_1, E_2 and E_3 ,

$$P(E_1/A) = \frac{P(E_1)P(A/E_1)}{P(E_1)P(A/E_1) + P(E_2)P(A/E_2) + P(E_3)P(A/E_3)}$$

$$P(E_2/A) = \frac{P(E_2)P(A/E_2)}{\text{Same}}$$

$$\text{gurukul } P(E_3/A) = \frac{P(E_3)P(A/E_3)}{\text{Same}}$$